An intercity Yb/Sr optical clock comparison at the 2.2×10^{-18} uncertainty over a 1 350 km fiber link

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Abstract

Redefining the International System of Units (SI) second requires independent frequency-ratio measurements between different types of optical clocks in different laboratories, with an overall measurement uncertainty below 5×10^{-18} and consistent results. However, although several optical clocks have reported self-evaluated uncertainties below 2×10^{-18}, validation of these self-assessed values through frequency-ratio comparisons has not yet been achieved. In this study, we compared a ytterbium optical lattice clock in Wuhan with a strontium optical lattice clock in Shanghai—approximately 700 km away—using a phase-stabilized fiber link spanning 1 350 km. The Yb/Sr frequency ratio is determined as 1.207 507 039 343 337 720 1(11), with a Bayesian statistical uncertainty of 1.14×10^{-18} and a total uncertainty of 2.2×10^{-18}. The value differs from that of NIST–JILA 1.207 507 039 343 337 723 0(37) by -2.87×10^{-18}, consistent within the margin of statistical error. To our knowledge, this is the first intercity, cross-species frequency-ratio measurement to meet the uncertainty requirements outlined in the roadmap for redefining the SI second. Furthermore, together with the NIST–JILA frequency-ratio result, this achievement satisfies the mandatory criteria (I.2.b) for the redefinition of the second, confirming the validity of uncertainty assessments for both ytterbium and strontium optical clocks.

1 Introduction

Optical lattice and single-ion clocks now reach fractional systematic uncertainties at or below 2×10^{-18} [1–9], two orders of magnitude better than current primary frequency standards [10;11]. Comparing independent clocks in different regions and laboratories verifies these self-assessed precision budgets more objectively: the environmental conditions and geodetic assumptions entering a self-evaluation are typically tied to a single location, and identical methods within one laboratory may share common systematic biases. Linking clocks of different types and locations into a collaborative network is therefore a key pathway toward establishing optical clocks as a verifiable metrological infrastructure [12;13]. The idea has moved from proposal to practice: fiber links with noise levels below those of the clocks themselves now span thousands of kilometers [14;15], and multi-clock campaigns have measured tens of ratios among clocks in six countries [16] and across a continental network [17]. Satellite-based advanced links with comparable performance [18–20] are also being developed and will interconnect with fiber networks to form a future global optical clock network.

The most immediate purpose of such a network is the redefinition of the SI second. One of the mandatory requirements (I.2.b) in the roadmap for this redefinition [21] is that frequency ratios of different clock types be measured at least twice by different institutes in agreement with an overall uncertainty of the comparison $\Delta\nu/\nu < 5 \times 10^{-18}$—a requirement reiterated in the 2025 CCTF progress report [22;23]. However, comparisons across independent institutions and different locations have reached at best a precision of 7.7×10^{-18} [17]. So far, no frequency-ratio measurements between clocks in different cities have satisfied this requirement [24].

Closing that gap matters beyond the definition itself. Chronometrically leveled clocks map the gravitational potential of the Earth at centimeter resolution [1;25;26], and clock networks have been proposed and used as sensors for dark matter [27–29]. All of these applications assume that remote

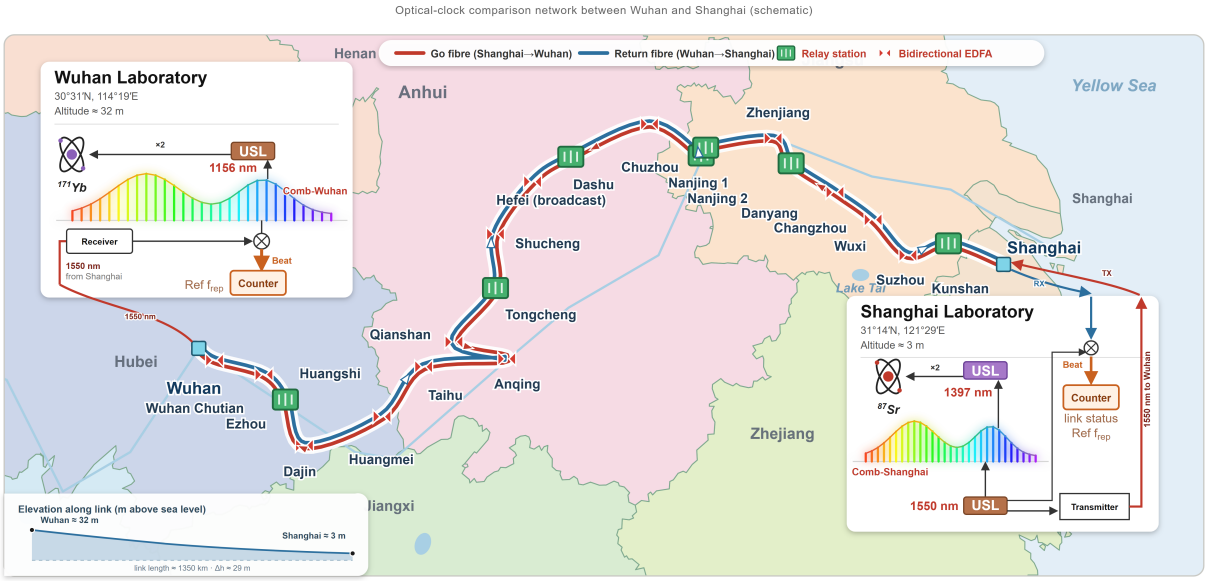


Figure 1: The two-clock network. An IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai are connected by a phase-stabilized 1 550 nm fiber link. Relay stations along the route carry erbium-doped fiber amplifiers and fiber-noise-cancellation loops.

clocks agree at a level that, until now, has been demonstrated only locally, in two experiments: a Sr/Yb/Al^+ triple-clock ratio measurement with an uncertainty of 3.2×10^{-18} by NIST–JILA in Boulder (US)[30], and a Yb^+/Sr measurement with 4.3×10^{-18} by PTB in Braunschweig (Germany)[31].

Here we report an intercity, cross-species frequency-ratio measurement between an IAPMST ytterbium lattice clock in Wuhan and a USTC strontium lattice clock in Shanghai, about 700 km apart, connected by a 1 350 km fiber link. We determine the Yb/Sr ratio with a total uncertainty of 2.2×10^{-18}; gravitational redshift effects have been corrected, whereas measurement error and the time-varying gravitational redshift term associated with gravitational tides have been retained as uncertainty components. The result establishes the two sites as a prototype node pair for a distributed optical-clock network, and completes, at intercity scale, the class of ratio measurements that the redefinition process depends on.

2 Results

2.1 Network and observation campaign

Figure 1 shows the two-clock network. An ytterbium (Yb) lattice clock at IAPMST[2] in Wuhan (systematic uncertainty 1.3×10^{-18}) and a strontium (Sr) lattice clock at USTC[4] in Shanghai (0.92×10^{-18}) are separated by 700 km and compared through a 1 350 km phase-stabilized fiber link (2 700 km loop-back) on a commercial telecom network used non-exclusively, with the carrier transmitted from Shanghai to Wuhan on the ITU-T 100 GHz DWDM channel C34 (1 550.12 nm, 193.4 THz). The link extends the field-deployed link of ref.[15], using the same cascaded fiber-noise-cancellation technique; its added instability stays below the clock noise for averaging times beyond 100 s.

Both combs are self-referenced and phase-locked to the local clock signal, and all other frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations (698/1397 nm on the Sr side; 578/1156 nm on the Yb side), so the two clocks are the system's only independent frequency sources. The recorded quantity is the out-of-loop beat between the received carrier and the local Yb-referenced comb, sampled every second and converted

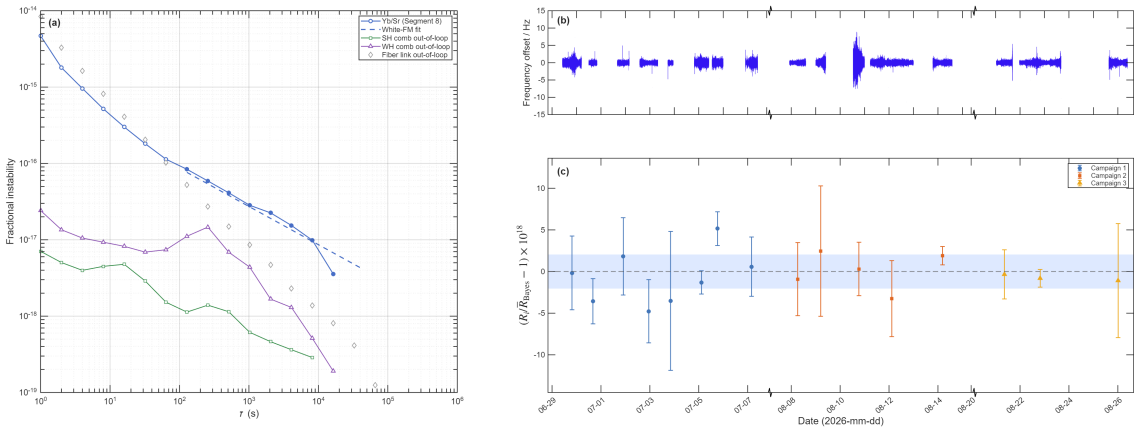


Figure 2: Frequency-ratio measurement results. (a) Fractional instability of a representative segment (Yb/Sr, segment 8); the points are the overlapping Allan deviation at different averaging times τ. The solid line is a white-frequency-noise model $\sigma_y(\tau) = \sigma_0/\sqrt{\tau/\mathrm{s}}$; the statistical uncertainty is the last computable point of the stability record, without extrapolation. The gray markers are the out-of-loop instabilities of the two frequency combs (SH, WH) and of the fiber link. (b) Per-campaign ratio measurements as fractional deviations from the final reported value, with error bars from the procedure in (a). (c) The same deviations by campaign date.

to fractional units by normalizing to the carrier frequency F_{1550} (1 550.12 nm); the fiber loop-back is monitored out of loop throughout. The geopotential difference between the two clocks was measured by spirit leveling to be $\Delta W = 280.042 \pm 0.085\ \mathrm{m^2\,s^{-2}}$ (see Methods). Each retained segment yields a frequency-ratio estimate through the fixed frequency-chain relations (see Methods).

2.2 Clock-ratio determination

Over two months (58 days) the two clocks and their transfer chains ran synchronously for a total of 274 effective hours, recorded as 16 jump-free segments. The results are presented in Figure 2.

In Figure 2a, the overlapping Allan deviation of the fractional ratio for a representative segment follows a white frequency noise law $\sigma_y(\tau) = \sigma_0/\sqrt{\tau/\mathrm{s}}$ over most of the record. The statistical uncertainty of each segment is obtained from the last computable point of its stability record rather than by extrapolating the white-noise fit to the segment duration; together with the segment mean, this provides one data point per segment, as plotted in Figure 2b,c.

We determine the Yb/Sr frequency ratio from the 16 segments. Each segment yields a ratio estimate with its own statistical uncertainty, after jump-point removal and endpoint screening (Methods). The segment estimates are consistent with their individual uncertainties, as quantified by the combination statistics in Table 1; 5 of the 16 segments have stability slopes that fall outside the white-frequency-noise band (groups 2, 6, 7, 12, 16). Following recent ratio comparisons[17;30], we combine the segments with a random-effect model that admits a common excess scatter[32;33], and evaluate three estimators (Table 1). The three agree at the 10^{-18} level; the Bayesian random-effect estimate, which carries the largest standard uncertainty, is reported as the primary value,

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1(11), \tag{1}$$

where the parenthesized uncertainty is the Bayesian statistical uncertainty. The total standard uncertainty of the comparison is 2.2×10^{-18} (Section 2.3).

Table 1: Statistical combination of the 16 segment ratios. WLS denotes weighted least squares with inverse-variance weights $w_i = 1/u_i^2$; ξ is the inferred common excess scatter. The weighted combination yields $\chi^2_{\mathrm{red}} = 0.412$ for 15 degrees of freedom ($p = 9.8 \times 10^{-1}$) and a Birge ratio of 0.642. All three estimators agree at the 10^{-18} level; the Bayesian random-effect value carries the largest standard uncertainty and is reported as the primary result.

Estimator	Yb/Sr	u (10^{-18})	ξ (10^{-18})
WLS (inverse variance)	1.207 507 039 343 337 720 130	1.128	—
Mandel–Paule	1.207 507 039 343 337 720 130	1.128	0.00
Bayesian random effect	1.207 507 039 343 337 720 134	1.139	0.14

Table 2: Uncertainty budget of the Yb/Sr frequency-ratio measurement. The time-varying gravitational-redshift term is carried as an uncertainty item and is not applied as a correction to the central value. Components are treated as independent.

Component	Contribution (10^{-18})
Statistical (Bayesian random effect)	1.139
Sr systematics	0.92
Yb systematics	1.3
Static gravitational potential	0.946
Fiber link	0.1
Frequency comb	0.1
Time-varying redshift	0.5
Total (quadrature)	2.24

2.3 Uncertainty budget

Table 2 collects the components of the total standard uncertainty. The statistical term dominates the budget, followed by the Yb and Sr systematic evaluations, the static-potential term from the leveling, and the time-varying-redshift term carried as an uncertainty item; the fiber-link and comb terms are negligible at this precision. The total standard uncertainty is 2.2×10^{-18}.

The time-varying-redshift term requires care, because the two clocks sample the geopotential at different times and places and no model of the temporal variation is applied to the reported value. Solid Earth and ocean tides, together with atmospheric, hydrological and non-tidal ocean loading, all change the local potential by amounts that reach the 10^{-18} level in fractional frequency over the campaign[34–36], and no single tidal model accounts for all of them consistently. Attempting to correct the beat record with the supplied tide series makes the segment scatter worse rather than better: the reduced chi-square of the combination rises from 0.41 to 0.57 (fixed theoretical amplitude) or 0.54 (fixed historical amplitude) when the template is applied before the ratio inversion. This is the signature of an inaccurate or incomplete geophysical model—the fit with the data is not consistent—so correcting the central value is not defensible at this stage. We therefore treat the time-varying gravitational redshift as an uncertainty item: its magnitude is set by the scale of the change that such a correction would induce, 0.85×10^{-18} in the central value, which we round up and carry as 0.5×10^{-18}. A physically complete, data-consistent treatment of time-dependent gravity is deferred to future work; the static potential difference from leveling remains the applied correction (Methods).

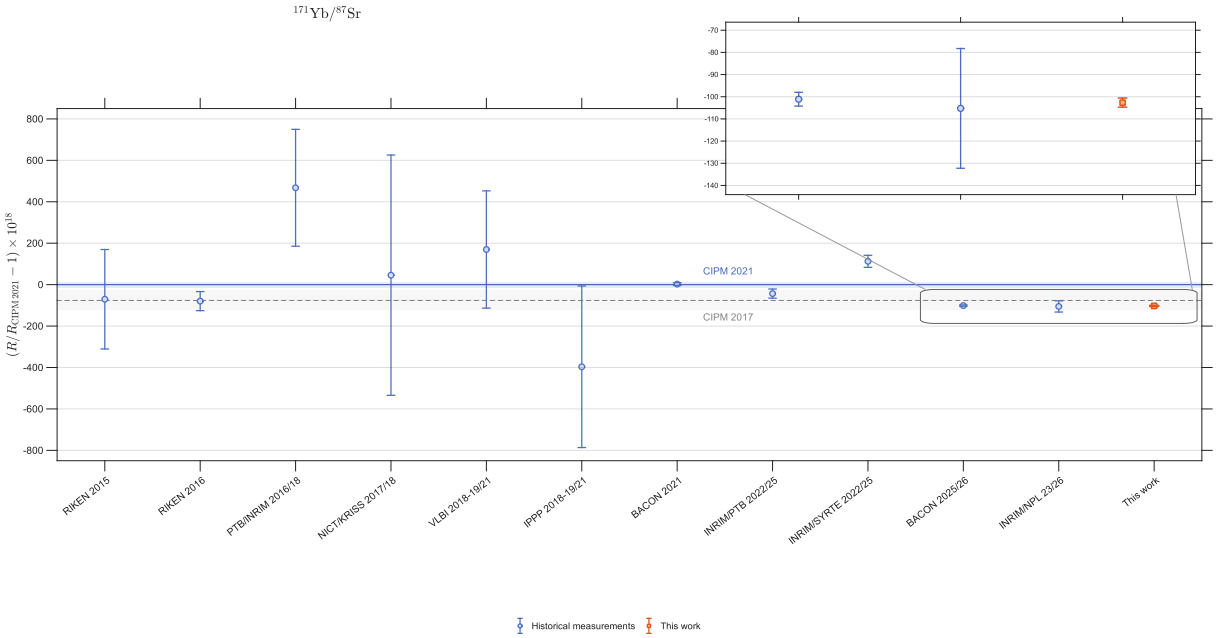


Figure 3: Evolution of the Yb/Sr frequency-ratio measurements. Deviations from the final value are shown against the measurement year for the historical determinations and for this work. The successive measurements converge toward agreement at the few-10^{-18} level.

3 Discussion and outlook

The value reported here agrees with the two existing measurements of the Yb/Sr ratio at the level set by the redefinition roadmap (Figure 3). Relative to the NIST–JILA value $1.207\,507\,039\,343\,337\,723\,0(37)$ [30], our central value differs by -2.87×10^{-18}; relative to the European-network value $1.207\,507\,039\,343\,337\,718(32)$ [17], it differs by $+2.13\times10^{-18}$. Both differences lie within the 5×10^{-18} agreement threshold of the roadmap's frequency-ratio criterion[21;23]. To our knowledge, this is the first reported intercity (100-km-scale), cross-species frequency-ratio measurement meeting that uncertainty requirement (as of *October* 2026); existing ratios at or below the threshold were measured on a single campus[30], while inter-institute comparisons through fiber networks have so far reached 7.7×10^{-18} [17]. The result closes, at intercity scale, the class of frequency-ratio measurements that the SI-second redefinition process depends on. Meeting the roadmap's frequency-ratio requirement has so far required clocks on a shared campus, where a single stabilized link serves both systems; here the two clocks sit in different cities, in different institutions, and use different atomic species, and the measurement still reaches the threshold, because the fiber link and the two combs contribute well below the clocks themselves. The bottleneck for reaching the redefinition threshold is therefore no longer the transfer infrastructure but the systematic evaluations of the clocks and the geodetic determination of their potential difference.

The three independent Yb/Sr values also track the progress of the comparison technology itself. The Yb/Sr ratio was first measured in 2014[37] and refined to 5×10^{-17} [38]; the BACON campus network then reached fractional ratio uncertainties at or below 8×10^{-18} [29], followed by 7.7×10^{-18} across a continental fiber network and 3.2×10^{-18} on a single campus. Our measurement continues this sequence at intercity scale with 2.2×10^{-18}, and the three latest values agree with one another at the few-10^{-18} level. The comparability that matters for the redefinition is no longer a single laboratory's claim about its own clock: it is the agreement between independent measurements, and the convergence seen here is what converts self-assessed systematics into verified ones.

Changes in the Earth's gravitational potential enter the comparison directly. The static difference between the two clock transitions is fixed by first-order spirit leveling and is carried

in the uncertainty budget; of the time-varying contributions, the tidal term is carried as a data-driven uncertainty item, while co-seismic steps from the regional seismicity during the campaign fall well below the reported precision. At this accuracy, it is the geodetic determination of the potential difference, rather than its variability, that limits the comparison; the same sensitivity underlies proposed chronometric-geodesy applications of optical-clock networks[1;25;26].

The treatment of the time-varying term illustrates this point. Applying the supplied tide series as a template correction leaves the residual segment scatter larger, not smaller (Section 2.3): a tidal model that is adequate for geophysical purposes is not necessarily adequate at the 10^{-18} level, where ocean-tide loading, hydrological mass redistribution and their local phase and amplitude errors all matter[35]. A correct time-dependent treatment would require a dedicated geodetic campaign with in-situ gravimetry and site-specific loading models; that is beyond the scope of this paper and is left to future work. Until such a treatment exists, the reported ratio carries the time-varying term conservatively in its uncertainty budget.

The campaign also demonstrates the operational character that a future optical-clock network will need. Over the two-month campaign the two clocks were restarted, the transfer chain relocked, and the beat record interrupted many times; the segment-level ratio estimates show no evidence of a systematic offset between the three measuring stages, so the reproducibility of the comparison does not depend on continuous operation of either system. This is the practical counterpart of the coordinated multi-clock campaigns[16;17]: the network tolerates interruption, provided each retained window is self-consistently characterized.

Several limitations bound the present result. First, the per-segment statistical uncertainties taken from the last computable Allan-deviation point are the dominant budget term, and five of the sixteen segments carry model-limited slope diagnostics; the random-effect treatment accounts for any remaining excess scatter at the level of the reported uncertainty, but a longer record would reduce the statistical term. Second, the systematic evaluations of the two clocks are those published by the laboratories; independent verification of those budgets, as opposed to the transfer, is not attempted here. Third, the geopotential difference is taken from a first-order leveling campaign; an alternative value exists in the experiment records and is carried as an external item pending reconciliation, as is the physical-model provenance of the supplied tide series and the exact accounting of the full sample totals. Finally, the present campaign meets the uncertainty requirement of the roadmap but not the full criterion, which additionally requires each ratio to be measured at least twice by different institutes and repeated over an extended period.

Looking ahead, the result establishes this link as a prototype node pair of a distributed optical-clock network: the same infrastructure can be extended to further nodes, and the technique demonstrated here – clocks at different sites agreeing at the redefinition threshold – is the operation that both the redefinition and post-redefinition applications require. The building blocks for such a network are now in place: terrestrial fiber links of the kind used here[14;15] carry comparisons over thousands of kilometers within continents, while free-space and satellite-based links are closing the remaining gaps[18–20;39]. The natural next step is a hybrid architecture in which optical-clock nodes on continental fiber backbones are interconnected across oceans by satellite or free-space optical links, so that any pair of clocks on Earth can be compared at the level demonstrated here. Realizing it will put the same demands on geodetic infrastructure that this comparison faced[1;25].

4 Conclusion

We have compared an ytterbium lattice clock in Wuhan and a strontium lattice clock in Shanghai over a 1 350 km fiber link and determined the Yb/Sr frequency ratio with a total standard uncertainty of 2.2×10^{-18}, the first intercity, cross-species ratio measurement to meet the uncertainty requirement of the roadmap for the redefinition of the SI second. Together with the

published same-campus and continental-network ratios, the three independent determinations now agree at the few-10^{-18} level. The comparison shows that the transfer infrastructure is no longer the bottleneck; what remains are the systematic evaluations of the clocks and the geodetic treatment of the gravitational potential, including a time-varying term that the available tide models cannot yet correct. With these handled, the two sites become a working node pair of a global optical-clock network built from fiber backbones joined by satellite and free-space links.

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Methods

Leveling determination of the geopotential difference

The static gravitational-redshift correction is fixed by the geopotential difference between the two clocks' atomic transition positions, determined by first-order spirit leveling together with simultaneous gravimetry. A leveling line of about 1 113 km was run between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC) through Ezhou, Huangshi, Huanggang, Lu'an, Hefei, Wuhu, Guangde and Suzhou, using digital levels with sight lengths below 30 m and gravity observed at every leveling point with two relative gravimeters. The orthometric height difference between the benchmarks follows from the leveled normal

heights after the normal-to-orthometric conversion, and the geopotential difference is integrated segment by segment, $\Delta W_{AB} = -\sum_i g_i\,\Delta H_i$, where g_i is the mean gravity and ΔH_i the orthometric height difference of the i-th segment. Combining this with the two local offsets between benchmark and clock platform (obtained by trigonometric leveling) and the platform-to-transition offsets from the clock design parameters gives the atomic-transition geopotential difference $\Delta W = 280.042 \pm 0.085\ \mathrm{m^2\,s^{-2}}$, equivalent to an optical fractional shift of -3.116×10^{-15}. The per-kilometer misclosure of the leveling is 0.255 mm, below the first-order limit.

Experimental setup and frequency transfer

The comparison links two optical lattice clocks at different institutions: an ytterbium (Yb) clock at IAPMST in Wuhan and a strontium (Sr) clock at USTC in Shanghai, about 700 km apart. The two sites are connected by a phase-stabilized fiber link of 1 350 km single-pass length (2 700 km loop-back), operated at 1 550 nm. The route is divided into spans of tens of kilometers, each terminated by a relay station that carries a narrow-linewidth laser together with an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; the added instability of the link is kept below the clock noise level for averaging times beyond 100 s.

Both ends use independent optical frequency combs to bridge the spectral gap between the clock transitions and the telecom carrier. On the Shanghai side the Sr 698 nm transition is transferred via a 1 397 nm continuous laser to the 1 550 nm carrier; on the Wuhan side the received carrier is linked to the Yb 578 nm transition through the 1 156 nm light. All local frequency references and phase-locked loops are derived from the comb repetition rate $f_{\mathrm{rep}} = 200\,\mathrm{MHz}$. The observable is the out-of-loop beat between the received carrier and the local Yb-referenced comb, converted to fractional units with the transfer frequency $F_{1550} = 193.3992\,\mathrm{THz}$. All frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations, $f^{1397}_{\mathrm{comb}} = \frac{1}{2}(1+a)\,f_{\mathrm{Sr}}$ and $f^{1156}_{\mathrm{comb}} = \frac{1}{2}(1+b)\,f_{\mathrm{Yb}}$, where a and b collect the Sr and Yb systematic corrections (density, lattice, second-order Zeeman and blackbody shifts). The static gravitational-redshift correction enters the ratio inversion as a multiplicative factor, $(\mathrm{Yb/Sr})_{\mathrm{corr}} = (\mathrm{Yb/Sr})_{\mathrm{raw}} \times (1+\delta_g)$ with $\delta_g = -3.116 \times 10^{-15}$, derived from the leveled geopotential difference above.

Data processing and segment definition

The counters sample the beat every second in phase-averaging mode. Samples deviating by more than 10 Hz from the running median are treated as jump points and removed; the retained data are further trimmed at both ends by an endpoint screen that drops points beyond 1% of the peak-to-peak range of the record. Each accepted segment contributes its longest jump-free record after this screening. Table 3 lists the per-segment retained counts, ratio deviations and statistical uncertainties; the category column marks the five segments whose stability fits are model-limited.

Clock-ratio determination and statistical model

Each retained segment yields a clock-ratio estimate R_i from the mean of its retained beat samples, inverted through the fixed frequency-chain relations and corrected for the static gravitational redshift $\delta_g = -3.116 \times 10^{-15}$ derived from the leveled geopotential difference (see above). The per-segment statistical uncertainty u_i is obtained from the overlapping Allan deviation of the beat fractional frequency, evaluated on a tau grid up to a quarter of the longest gap-free run T_i of the segment: u_i adopts the last computable point $u_i = R_i\,\sigma_y(\tau_{\max})$ without extrapolating the stability curve beyond the data. The log-log slope of σ_y against τ is computed as a diagnostic; segments whose slope leaves the white-frequency-noise band are flagged but retained.

The segment estimates are then combined under a random-effect model with a common excess scatter[32;33]: each segment is modeled as $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$, with ξ added in quadrature

Table 3: Per-segment data of the 16 segments analyzed in this work. Ratio deviations $y_i = R_i/R_1 - 1$ are relative to the first segment; $u(y_i)$ is the statistical uncertainty of the segment ratio; "Fit" is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

Group	n_{valid}	y_i (10^{-18})	$u(y_i)$ (10^{-18})	Fit
1	68 009	+0.00	4.245	est.
2	29 481	-3.36	3.017	lim.
3	42 891	+2.00	5.880	est.
4	64 280	-4.61	3.674	est.
5	19 532	-3.38	15.250	est.
6	52 140	-1.15	2.230	lim.
7	39 526	+5.31	3.476	lim.
8	43 565	+0.97	9.860	est.
10	57 938	-0.76	5.003	est.
11	35 824	+2.61	5.976	est.
12	40 346	+0.73	4.843	lim.
13	152 994	-3.09	4.019	est.
14	68 618	+1.97	6.254	est.
15	57 649	-0.25	4.028	est.
16	147 640	-0.77	1.809	lim.
17	65 983	-0.89	8.813	est.

to the per-segment uncertainties. The primary value is obtained from a Bayesian random-effect model, in which μ and ξ are jointly inferred with flat priors on μ and on $\log \xi$, and the joint posterior $p(\mu, \xi \mid \{R_i, u_i\})$ is marginalized numerically over a 40-decade $\log \xi$ grid; the posterior mean of μ and its posterior standard deviation give the reported central value and statistical uncertainty, with $\xi = 0.14 \times 10^{-18}$. As cross-checks, we also combine the segments by weighted least squares (inverse-variance weights $w_i = 1/u_i^2$), which yields $\chi^2_{\mathrm{red}} = 0.412$ for 15 degrees of freedom ($p = 9.8 \times 10^{-1}$), and by the Mandel–Paule estimator[33], which gives $\xi = 0.00 \times 10^{-18}$; the Birge ratio is 0.642. The excess scatter quantified here also absorbs the correlated component of the time-varying gravitational-redshift term (see below), which is carried separately as an uncertainty item rather than corrected.

Uncertainty budget

Table 2 lists the uncertainty components of the Yb/Sr ratio. The statistical term is the posterior standard deviation of the Bayesian random-effect model applied to the 16-segment data. The Sr and Yb systematic terms are the published evaluations of the two clocks[2;4]; the value adopted for the Yb clock resolves an internal-versus-published discrepancy in favor of the published evaluation. The static-potential term is $u(\Delta W)/c^2$ from the leveling uncertainty 0.085 $\mathrm{m^2\,s^{-2}}$, giving 0.946×10^{-18}. The fiber-link and comb terms are experiment-side upper bounds of 1×10^{-19} each, carried at 0.1×10^{-18} and 0.1×10^{-18} to preserve the experiment-side margin.

The time-varying gravitational-redshift term is carried as an uncertainty item, quantified from the scale over which the segment-averaged shift varies across the campaign, and is not applied as a correction to the reported central value. As a direct sensitivity check, we repeated the combination with the supplied tide series applied as a template correction before the ratio inversion, at fixed theoretical and at fixed historical amplitudes: the reduced chi-square of the combination rose from 0.41 to 0.57 and 0.54, respectively, and the Bayesian center moved by up to 0.85×10^{-18}. A template that degrades the agreement between segments is treated as inconsistent with the data, so no tidal correction is applied; the magnitude of the induced center

shift is carried as the time-varying-redshift uncertainty item. The components are treated as independent and combined in quadrature, giving the total standard uncertainty 2.24×10^{-18}.

Data availability

The frozen, machine-readable result layer underlying this manuscript is archived in the repository (`https://github.com/hjiang555-a11y/gravitation-whsh`) under `results/audit-v1/verified/`: the manifest with SHA-256 `ac725aaba8ec744bf4e7a17cf37cee41af57461f0d3684584711b24a02b0962e`, the derived ledgers and statistics, and the generated audit report. The raw beat files and the leveled geopotential inputs are not redistributed.

Code availability

The analysis and manuscript pipelines are in the same repository under the Apache License 2.0. The manuscript numbers are generated by `paper/rev2/tools/build_numbers.py`, which reads only the frozen manifest and the curated constants listed in `generated/numbers_manifest.json`.